

ACTU2200 - Asset and Liability Management in Insurance

Academic Year 2019 - 2020

Assignment

April 4, 2020

1 Deterministic Model

We consider an insurance company with the portfolio of insurance contracts described in the excel file “Insurance Portfolio (2019-2020).xls”. The assets of this insurance company are supposed to be invested in the following (risk-free government) bonds with the given allocation:

- 25% in a coupon bond with maturity 2 years, coupon rate 2.00%;
- 45% in a coupon bond with maturity 5 years, coupon rate 2.50%;
- 30% is invested in a coupon bond of 8 years, coupon rate 3.50%.

The initial term structure of the continuously compounded (risk-free government) interest rates is described by a Nelson-Siegel model with the following parameters:

$$\begin{aligned}\beta_0 &= 0.015 \\ \beta_1 &= -0.018 \\ \beta_2 &= -0.015 \\ \tau &= 1.5\end{aligned}$$

Question 1: Compute the value of the statutory mathematical reserve of this insurance company at time 0, with and without premiums.

Question 2: Compute the economic value of the technical provisions of this insurance company at time 0, with and without the premiums. Compare these values to the statutory mathematical reserve of Question 1. Explain the difference.

Question 3: Compute the duration and the convexity of the assets and of the technical provisions as well as the duration gap and convexity gap of the company. Assume no premium is paid. For this, suppose the economic value of the equity is equal to 10% of the economic value of the technical provisions.

Question 4: Compute the duration and the convexity of the assets and of the technical provisions as well as the duration gap and convexity gap of the company, assuming this time that the premiums are paid. Compare your answers to those of Question 3. Comment and explain your results.

Rem: Assume here that the economic value of the equity is the same as in Question 3 (in other words, reduce the current value of the assets by an amount equal to the present value of the premiums).

Question 5: What would (approximately) be the impact of an instantaneous 1.5% upward parallel shift and of 1.5% downward parallel shift of the term structure on the economic value of the equity of this insurance company (with and without premiums). Check the accuracy of your answer by computing the exact impact of these shifts on the equity. Explain your results.

Question 6: You believe that the term structure of the continuously compounded rates will not decrease by more than 0.75% with a 98% probability. You accept the risk of losing a maximum of 10% on your equity with the same probability. What should be the duration of your asset portfolio?

Question 7: A standard interest rate swaps on Euribor 6 months with maturity 15 years is available. What position should you take in this swap to immunize your equity. Check the (exact) impact on the equity, of a 1% instantaneous upward and downward shifts of the term structure when such a swap is included. Compute the convexity gap taking into account your position in the swap. Explain your results.

Question 8: Compute the dollar key rate durations of the asset portfolio, of the technical provisions and the equity for the following key rates [0Y, 1Y, 5Y, 10Y, 20Y]. In each case, compute the sum of the key rate dollar durations.

Question 9: Compute the key rate dollar durations (for the key rates defined in the previous question) of a Forward Rate Agreement (FRA) on Euribor 3 months, with a maturity of 6 months and a nominal of 1million Euros.

Question 10: If you want to immunize your company to changes in the 1 year key rate, what position should you take in this FRA (what nominal? buyer or seller FRA ?)

Rem: Focus only on the key rate duration and neglect the convexity.

Question 11: Compute the key rate dollar durations (for the key rates [0Y, 1Y, 5Y, 10Y, 20Y]) of a swap on the Euribor 6 months, with maturity 15 years and a nominal of 100 millions. Assume the fixed leg makes annual payments.

Question 12: If you want to immunize your company to changes in the long term rate (here assumed to be the key rate 20Y), what position should you take in the swap described above?

Rem: Focus only on the key rate duration and neglect the convexity.

Remark:

To answer the questions above, please make the following assumptions:

1. The premiums are assumed to be paid at the beginning of the year. In particular, the premiums paid during the first year, are assumed to fall at time 0_+ . Therefore, in questions 1 and 2, the value of the technical provisions (statutory or economic) are assumed to be computed right before the payment of these premiums.
2. Whenever it is asked to take into account the premiums of the insurance contracts, please assume that the value of the equity remains the same as in the case where no premium is paid.
3. From question 7 to 13, consider only the case where the premiums are paid.
4. Assume the Euribor rates and the zero-coupon swap rates are identical to the risk-free government rates described above (i.e. the Nelson-Siegel model).
5. No expense, commission or tax are paid.
6. The mortality of the policyholders follows Makeham's model with the MR table parameters i.e.

$$\begin{aligned}
 k &= 1000266.63 \\
 s &= 0.999441703848 \\
 g &= 0.999733441115 \\
 c &= 1.101077536030
 \end{aligned}$$

2 Stochastic Model

2.1 Description

2.1.1 The Financial Market

We now assume that the financial market is made of a mutual fund, of a short term deposit and of zero-coupon bonds of all maturities. We assume that there is no arbitrage in this financial market. The value at time t , $B(t)$, of the short term deposit continuously increases at the instantaneous risk-free rate $r(t)$. This rate $r(t)$ is here assumed to be solution of the following stochastic differential equation:

$$dr(t) = a_r (\theta(t) - r(t)) dt + \sigma_r dW_1^Q(t)$$

where $W_1^Q(t)$ is a Brownian motion under the risk-neutral measure and where

$$\begin{aligned}
 a_r &= 0.15 \\
 \sigma_r &= 0.015
 \end{aligned}$$

The initial term structure is given by the Nelson-Siegel model described above.

The market value $S(t)$ of the mutual fund is given by the following stochastic differential equation:

$$dS(t) = r(t)S(t)dt + \sigma_S S(t)dW_2^Q(t)$$

where $W_2^Q(t)$ is a Brownian motion under the risk-neutral measure and where $\sigma_S = 0.15$. The two Brownian motions $W_1^Q(t)$ and $W_2^Q(t)$ have a correlation equal to $\rho = 0.20$.

Finally, the vector λ of market prices of risk is given by

$$\lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} -0.200 \\ 0.45 \end{pmatrix}$$

2.1.2 The Insurance Company

The value of the assets is the same as in the previous section (assume no premium is paid). The assets of the insurance company are now invested in the mutual fund described above as well as in a zero-coupon bond with maturity $T = 15$ years. The asset portfolio is supposed to be continuously rebalanced to the following fixed asset allocation:

$$x_{MutFunds} = 0.10\%$$

$$x_{bond} = 0.90\%$$

This means that the value of the assets $V^a(t)$ is given by the following stochastic differential equation:

$$dV^a(t) = x_{MutFunds} V^a(t) \frac{dS(t)}{S(t)} + x_{bond} V^a(t) \frac{dP(t, T)}{P(t, T)}$$

where $P(t, T)$ is the price at time t , of a zero-coupon bond with a time-to-maturity $(T - t)$.

On the liability side, we consider the same insurance portfolio as in the deterministic section. The insurance company shares with its policyholders a fraction of its profits at the end of the year. More precisely, on top of the guaranteed rate, the management distributes a profit-sharing rate at the end of year $k = 1, 2, \dots$, based on the following rule:

$$r_i^{PS}(k) = \min(b \max(r_A(k) - d - g_i, 0), 3\%)$$

where $b = 0.40$, $r_A(k)$ is the return of the asset portfolio over the time interval $[k - 1, k]$, $d = 0.005$, g_i is the technical rate for the policy i . Assume no profit sharing has been distributed the previous year (i.e. $r_i^{PS}(0) = 0$).

2.2 Valuation

Question 1:

1. What should theoretically be the values of the following mathematical expressions:

$$E^Q \left[\frac{1}{B(t)} \frac{S(t)}{S(0)} \middle| \mathcal{F}_0 \right]$$

and

$$E^Q \left[\frac{1}{B(t)} \frac{P(t, T)}{P(0, T)} \middle| \mathcal{F}_0 \right]$$

for $T \geq t$ and where $P(t, T)$ is the price at time t of a zero-coupon bond of maturity T .

2. Based on the outcomes of your Monte-Carlo simulation, estimate the values of these expressions for time $t = 1$, $t = 5$ and time $t = 10$ and for $T = 15$. Compare your estimates with their theoretical values.

Question 2: Give the solution of the stochastic differential equation of the value of the asset portfolio $V^a(t)$, for $t \in [0, 1[$.

Question 3: Compute (based on the outcomes of a Monte-Carlo simulation) the economic value of the technical provisions with and without profit-sharing. Compute the economic value of the profit-sharing. What is the impact of the profit-sharing on the economic value of the equity. Discuss your results.

Question 4: Compute the impact of a 1% downward parallel shift of the term structure on the value of the equity.

Question 5: Compute the effective duration of the value of technical provisions, based on an upward and downward parallel shifts of the term structure of 0.5%. Compare it the duration computed in Part I. Explain your results.

Question 6: Based on your answer to Question 5, compute the duration gap of your company. What is (approximately) the impact of a 1% downward parallel shift of the term structure on your equity. Discuss your result.

2.3 Risk Assessment

Assume in this Section that there is no profit-sharing.

Question 1: Compute the risk premium of the mutual fund and of a zero-coupon bond with time-to-maturity $(T - t) = 15$ years. Compute the risk premium of the asset portfolio.

Question 2: Compute and draw the probability density function of the gain on the asset portfolio. Give its mean and its standard deviation. Compute the V@R at 99.5%. Assume in this question that there is no liability.

Question 3: Same question as 2 but now take into account the impact of the liabilities. Explain the difference.

Question 4: Compute and draw the probability density function of the economic value of the technical provisions at the beginning of the next year. Compute its mean and standard deviation. Explain your calculations.

Question 5: Compute and draw the probability density function of the economic value of the equity at the end of the year. Compute its mean, its standard deviation and its 0.05% percentile.

Question 6: Compute the Solvency Capital Requirement (SCR) at a 99.5% confidence level. Compare the SCR with the current economic value of the equity. Discuss your results. Explain the assumptions you used in the computation of the SCR.

Question 7: Compute the expression of the diffusion term of the economic value of the technical provisions. Compare it to the diffusion term of the value of the asset portfolio. Comment.

Question 8: Based on your previous answer, what could you do to mitigate the interest rate risk of the company?

3 Instructions

Do not forget to check that the number of simulations is sufficient for your results to be stable.